

IBM BOB AI INNOVATION HACKATHON 2026 • TRACK: AI • PROBLEM U1



Grid Failure Advisor

Predicts outage-prone grid assets weeks in advance by fusing sensor telemetry, weather forecasts, and historical incidents into one explainable, prioritized maintenance plan.

Team: Grid Failure Advisors

Utilities find out too late — after the outage, not weeks before it.

Calendar-based maintenance replaces parts on a fixed schedule, regardless of actual condition — wasting money on healthy assets while assets that degrade between cycles fail without warning.

- Sensors already measure **temperature, vibration, partial discharge, and oil quality** — and show failure signatures **weeks in advance**.
- Weather forecasts show which regions face compounding stress — but are **never combined** with sensor data in time to act.
- A single major transformer/substation failure strands tens of thousands of customers, including hospitals and water treatment plants.

\$1M+

PER HOUR, COST OF A MAJOR BLACKOUT

weeks

OF ADVANCE WARNING ALREADY SITTING IN UNUSED
SENSOR DATA

One explainable score. One prioritized, weather-timed plan.

Grid Failure Advisor fuses sensor data, weather forecasts, and incident history into a composite risk score where **every point traces to a named, human-readable factor** — not a black box.

- Ranks assets by **risk × grid impact severity**, not risk alone — a hospital feeder outranks a low-consequence asset at equal risk.
- Pulls maintenance dates **forward, ahead of a forecast storm** in the asset's region.
- A **Grid Copilot** answers natural-language questions, grounded in the same real data.

Sensors + Weather + History



risk_engine.py — explainable score



maintenance_planner.py — ranked, weather-timed plan



Dashboard + Grid Copilot + MCP Server

Every asset ranked, every score explained.

■ GRID FAILURE ADVISOR / control console

DASHBOARD

MAINTENANCE PLAN

ASSET RISK OVERVIEW

All monitored grid assets, ranked by predicted failure risk.

Region

ALL

Risk Tier

ALL

50 of 50 assets

NAME	TYPE	REGION	TIER	Risk Score ↓	GRID IMPACT	CUSTOMERS
Eastgate Transformer 04	transformer	Eastgate	■ CRITICAL	100	7	14,612
North Valley Substation 11	substation	North Valley	■ CRITICAL	100	8	48,708
Eastgate Transformer 03	transformer	Eastgate	■ CRITICAL	97.6	7	8,453
North Valley Transformer 04	transformer	North Valley	■ CRITICAL	92.6	7	6,477
Eastgate Transformer 02	transformer	Eastgate	■ CRITICAL	90.5	7	4,108
Highland Substation 03	substation	Highland	■ CRITICAL	84.4	8	41,031
Bayview Transformer 05	transformer	Bayview	■ CRITICAL	76.1	6	13,616
Bayview Transformer 03	transformer	Bayview	■ HIGH	74	6	11,032

COPILLOT

The top-ranked asset: a real degrading sensor trend and an incoming storm — verified live, not staged.

■ GRID FAILURE ADVISOR / control console

DASHBOARDMAINTENANCE PLAN

← Back to dashboard

Eastgate Transformer 04

■ CRITICAL

100/100

TYPE	REGION	INSTALLED
transformer	Eastgate	1989
CAPACITY	CUSTOMERS SERVED	GRID IMPACT
24.9 MVA	14,612	7/10

WHY THIS SCORE

+25 ptsWeather Risk

Storm warning forecast for Eastgate on 2026-09-18: wind 57.9 mph, precipitation probability 81% — compounds any existing equipment stress in this region.

+18 ptsSensor Anomaly — Partial Discharge

Partial Discharge has risen 63% versus its 15-day baseline (81.4 → 133.0), consistent with developing equipment stress.

+18 ptsSensor Anomaly — Oil Quality

Oil Quality has fallen 36% versus its 15-day baseline (84.9 → 54.6), consistent with developing equipment stress.

+16.5 ptsHistorical Incident Rate

Transformers of similar age (37 yrs, 'old' bracket) show 8 historical incidents in the record; base failure rate estimated at 7.4%/yr from age and asset class.

+10 ptsSensor Anomaly — Vibration

Vibration has risen 15% versus its 15-day baseline (0.05 → 0.0575), consistent with developing equipment stress.

COPILLOT

Bob doesn't just build it. Bob can run it.

IBM Bob was used as the AI coding agent throughout the build — and the running app exposes an **MCP server** so Bob (or any MCP client) can call the exact same tools that power the app itself.

- **ONE IMPLEMENTATION**
the Grid Copilot's /copilot/ask and the standalone MCP server both call `grid_tools.py` — no separate "demo path."
- A judge can start `python -m app.mcp.server`, call `get_at_risk_assets(region="Eastgate")`, and get back the **exact numbers** the dashboard shows.
- 5 tools: `get_at_risk_assets`, `get_asset_detail`, `explain_asset_risk`, `get_maintenance_plan`, `list_regions`

The screenshot displays two side-by-side panels from a web application. The left panel, titled 'GRID FAILURE ADVISOR / control console', shows a 'MAINTENANCE PLAN' generated on 15/09/2026 at 05:30:00. It lists several assets in the 'NORTH VALLEY' region, each with a risk score and a status (CRITICAL, HIGH, or MEDIUM). The right panel, titled 'GRID COPILOT', shows a query 'which assets are highest risk near Eastgate?' and a response listing the top 4 highest-risk assets in Eastgate, including Eastgate Transformer 04, Eastgate Transformer 03, Eastgate Transformer 02, and Eastgate Substation 01. A 'Tools called (1)' button is visible. At the bottom of the right panel, there is a text input field with the placeholder 'Ask the grid copilot...' and a 'Send' button. A red box in the bottom right corner of the right panel indicates 'BY 2026-09-16 id remaining'.

Beyond the hackathon.

- **Swap synthetic for real data** — the service-layer interface (`grid_tools.py`) doesn't change; the risk engine and MCP server both keep working.
- **Upgrade to a trained ML model** once real historical failure labels exist — designed for this from day one.
- **Add a real time-series database** behind the same `data_loader.py` interface for a production-scale fleet.
- **Extend the MCP surface** so Bob can not just query risk, but draft work orders and crew dispatch directly.

50

ASSETS MONITORED IN THIS DEMO — THE ARCHITECTURE
SCALES TO THOUSANDS